

● MEDIUM

● INFORMATION AND IDEAS

● ANSWER KEY

Information and Ideas

05

10 answers · Rationale-rich · Teach from doc

Q1**A**

- A. Researchers have obtained the most definitive evidence to date of the existence of the Wigner crystal.
-

Choice A is the best answer because it most accurately states the main idea of the text. According to the text, Eugene Wigner hypothesized that a crystal could exist that would be composed of electrons and have a honeycomb-like shape. The text goes on to say that the existence of the Wigner crystal remained unconfirmed until Feng Wang and colleagues were able to make an impression of one using two semiconductors and an ultrathin sheet of graphene. Thus, the main idea is that researchers have obtained the most definitive evidence to date of the existence of the Wigner crystal.

Choice B is incorrect because the text focuses on one kind of crystal—the Wigner crystal— and doesn’t discuss crystalline structures in general. And although the text conveys that Wang and colleagues figured out a way to capture an image of a Wigner crystal, it doesn’t address the idea of applying this approach to other types of crystals. Choice C is incorrect because the text describes in general the process Wang and colleagues followed to obtain an impression of the Wigner crystal; it doesn’t address the relative importance of each component in that process. Choice D is incorrect because the text doesn’t state that researchers had a hard time getting an impression of the Wigner crystal because of its honeycomb structure. Nothing in the text indicates why it took so long to prove the existence of this crystal or take an impression of it.

Q2

B

- B. has been preserved in certain mammal species because it benefits their immune systems.
-

Choice B is the best answer because it most logically completes the text's discussion of Smith and colleagues' investigation of the evolution and biological role of the appendix. The text indicates that the team found several instances of the appendix emerging and not disappearing in the lineages of various mammal species the team examined. Furthermore, the text states that species that possess an appendix also tend to have relatively high amounts of lymphoid tissue—a type of tissue that supports immune system function. Taken together, these details strongly support the hypothesis that the appendix has persisted in some species because it has a function that contributes to effective immune responses in those species.

Choice A is incorrect because the text doesn't address any nonmammalian species. Choice C is incorrect because the text doesn't make predictions about the evolutionary future of the species Smith and colleagues examined, and although the implication of the text is that the appendix likely does serve a function for the immune system, nothing in the text indicates that the appendix will become more widespread in the future. Choice D is incorrect. Although the text does suggest an association between having an appendix and relatively high concentrations of lymphoid tissue, it doesn't claim that the appendix causes the tissue to grow, nor does it address the relative production of the tissue at different periods of time.

Q3

B

- B. Waterbird communities at Old Road, a site with a relatively high percentage of shoreline consisting of bulkheads, had lower average IWCI scores than did waterbird communities at Miles, a site with a relatively high percentage of shoreline consisting of riprap.
-

Choice B is the best answer because it presents a finding that, if true, would most directly illustrate the researchers' finding about waterbird responses to shoreline hardening structures. The text explains that using the IWCI, an index on which higher scores indicate higher community integrity, researchers found that bulkheads are more strongly negatively correlated with waterbird community integrity than riprap is—that is, that bulkheads reduce the integrity of waterbird communities more than riprap does. The finding that waterbird communities at a site with a relatively high percentage of bulkheads along the shoreline (Old Road) had lower average IWCI scores than waterbird communities at a site with a relatively high percentage of riprap along the shoreline (Miles) did would illustrate the researchers' finding, since it would be an example of lower waterbird community integrity in a location with substantial bulkhead presence than in a location with substantial riprap presence.

Choice A is incorrect because a finding of similar waterbird community integrity in two sites predominated by bulkheads wouldn't reveal anything about waterbird community integrity at sites predominated by riprap or how waterbird community integrity compares between the two types of sites. Thus, the finding wouldn't illustrate the discovery of a difference in how bulkheads and riprap correlate with waterbird community integrity. Choice C is incorrect because it presents a finding that compares waterbird community integrity at sites with different overall amounts of shoreline hardening but without distinguishing between bulkheads and riprap. Such a distinction would be necessary to illustrate the researchers' finding of a difference in how bulkheads and riprap each correlate with waterbird community integrity. Choice D is incorrect because it presents a finding that addresses a difference in waterbird community integrity between one site with equal percentages of bulkheads and riprap along its shoreline (Curtis) and one site with unequal percentages of bulkheads and riprap (Miles) but without indicating whether bulkheads or riprap were more prevalent at Miles. This finding wouldn't provide any clear information about how bulkheads and riprap each correlate with waterbird community integrity, so it wouldn't illustrate the researchers' finding of a difference between those correlations.

Q4

C

- C. A person appreciates the extensive time he passed in the company of someone he deeply admired.
-

Choice C is the best answer because it most accurately describes the main idea of the text: a person appreciates the extensive time he passed in the company of a man he deeply admired. The speaker says that he spent "years" (a long period of time) in the company of a "truly great man" with "a natural genius for invention." These phrases indicate the speaker's feelings of admiration.

Choice A is incorrect because though the text indicates that the speaker feels professional admiration for the "truly great man," the text does not indicate that the speaker feels love for him; nor does the text indicate that the speaker's humility, absurdity, and slovenly nature proved to be an obstacle to this professional relationship. Choice B is incorrect because though the speaker describes himself as humble, the text does not indicate that he is exceptionally talented; nor does the text indicate that the "truly great man" professes humility or is concerned with whether he intimidates others. Choice D is incorrect because the text does not indicate how either the speaker or the "truly great man" express their ideas.

Q5

D

D. the timing of downbeats and offbeats may play a crucial role in giving a song swing.

Choice D is the best answer. The passage tells us that participants were “more than seven times likelier to characterize the modified songs as having swing than to characterize the original versions as having swing.” Because the modified songs had been changed by altering the timing of the downbeats and offbeats, this suggests that the timing of downbeats and offbeats may play a crucial role in giving a song swing.

Choice A is incorrect. Although the passage mentions that the researchers “synchronized the offbeats” in the modified songs, they also “delayed the downbeats” in those songs. Because we can’t disentangle whether it was the synchronized offbeats, the delayed downbeats, or the combination of both that increased the song’s swing, we don’t have enough information to make this inference. Choice B is incorrect. The passage doesn’t mention whether or not it’s difficult for a jazz musician to give a song swing, so there’s no basis for this inference. Choice C is incorrect. The passage doesn’t compare jazz songs that feature piano to those that don’t, so there’s no basis for this inference.

Q6

C

- C. “Forget not to care for the cat, the birds, and the flowers. Do not eat too quickly nor fan too vigorously now that the weather is warming.”
-

Choice C is the best answer because it presents a quotation that illustrates the claim that Mrs. Spring Fragrance demonstrates concern for what’s happening at home while she’s in California. By giving reminders to “care for the cat, the birds, and the flowers,” “not eat too quickly,” and avoid engaging in strenuous activity in the heat, Mrs. Spring Fragrance shows that she’s thinking about what’s happening at home and wants to ensure everything is taken care of.

Choice A is incorrect because the quotation, while it does suggest that Mrs. Spring Fragrance has made fudge at home before, is focused on preparations for an upcoming festival, not on concerns for anything happening at home while Mrs. Spring Fragrance is away. Choice B is incorrect because the quotation has to do with an upcoming event during Mrs. Spring Fragrance’s trip—visiting San José and meeting someone new—rather than her concern for what’s happening at home. Choice D is incorrect because the quotation is focused on how Mrs. Spring Fragrance feels about her trip and the friends she’s seeing, not on her concern for what’s happening at home.

Q7

D

- D. Although impractical, the design of the apartment building by Gins and Arakawa may improve the well-being of the building's residents.
-

Choice D is the best answer because it most accurately states the main idea of the text.

According to the text, conceptual artists Gins and Arakawa have designed an apartment building that is disorienting because of several unconventional elements, such as uneven kitchen counters and “a door to nowhere.” The text goes on to suggest that there may be benefits to this kind of design because filmmaker Yamaoka lived in the apartment building for four years and reported health benefits. Thus, although the design is impractical, it may improve the well-being of the apartment building's residents.

Choice A is incorrect. Although the text mentions that Yamaoka lived in the apartment for four years, it doesn't address how long someone can beneficially live in a home surrounded by fanciful features or whether doing so can be sustained. Choice B is incorrect. Although the text mentions the potential benefits of living in a home with disorienting design features, it doesn't suggest that this is the most effective method to create a physically stimulating environment. Choice C is incorrect because the text refers to Yamaoka to support the claim that Gins and Arakawa's apartment building design may be beneficial, but the text doesn't indicate that Yamaoka supports the designs of other conceptual artists.

Q8

A

- A. make themselves appear brave and strong for being able to climb to difficult places where edelweiss supposedly grew.
-

Choice A is the best answer because it most logically completes the text's discussion of mountain climbers' motivations for exaggerating the difficulty of reaching edelweiss. The text establishes that edelweiss became symbols of strength and courage in Switzerland in the mid-1800s. The text goes on to state that mountain climbing became popular around the same time and that climbers spread the false idea that the flowers grew only in dangerous, hard-to-reach locations. According to the text, historian Tobias Scheidegger argues that these claims were "self-interested," meaning they served the climbers' own purposes. The text therefore implies that by claiming that edelweiss could be found only in perilous locations, mountain climbers could enhance their own reputations: climbers who encountered edelweiss would appear to have undertaken a dangerous journey, thereby making themselves seem brave and strong.

Choice B is incorrect because making edelweiss seem more dangerous to reach than other mountain flowers are would likely discourage flower enthusiasts from trying to reach edelweiss rather than encourage them to explore the Swiss Alps. The text suggests that climbers' self-interested motivation for exaggerating the difficulty of climbing to areas where edelweiss grew was to enhance their own reputations for being especially brave, not to promote tourism in the Swiss Alps. Choice C is incorrect because the text states that according to Scheidegger, mountain climbers exaggerated the difficulty of reaching edelweiss for self-interested reasons, not because they wanted to report genuine scientific observations about the flower's unique characteristics. The text provides no indication that the climbers sincerely believed that their claims about the especially challenging terrain in which edelweiss grew were accurate, much less that they wanted to share this information with scientists. Choice D is incorrect because the text doesn't suggest that mountain climbers were trying to prove that edelweiss were more common in Switzerland than elsewhere. The text's focus is on the supposed difficulty and danger of reaching the flowers, not on their relative abundance in the Swiss Alps compared to other regions. In fact, the text implies that mountain climbers exaggerated how difficult edelweiss were to reach; it therefore wouldn't make sense to say that they exaggerated this difficulty to demonstrate that edelweiss were relatively common.

Q9

B

- B. In every year of the experiment, plants treated with the nitrogen fertilizer had a greater yield than did plants treated with the control fertilizer.
-

Choice B is the best answer because it describes data from the graph that support the researchers' conclusion that using nitrogen fertilizer increases cantaloupe production. The bar graph shows the cantaloupe yield for plants grown with nitrogen fertilizer and for those grown with a control fertilizer (without nitrogen) for three consecutive years (2017–2019). For each year in the graph, the yield for the nitrogen group is greater than the yield for the control group. In pounds per acre, the value in 2017 for the nitrogen-group yield is greater than 30 while the control-group yield is less than 25; in 2018 the nitrogen-group yield is greater than 40 while the control-group yield is less than 35; and in 2019 the nitrogen-group yield is greater than 25 while the control-group yield is less than 20. Thus, these data from the graph strongly support the conclusion that using nitrogen fertilizer increases cantaloupe yield.

Choice A is incorrect because the bar graph shows that in 2019 the nitrogen group had a yield below 30 pounds per acre. Choice C is incorrect. Although this choice accurately describes data in the bar graph—in 2018, the control-group yield is greater than 30 pounds per acre, and in 2019 the nitrogen-group yield is less than 30 pounds per acre—a claim that the control-group yield exceeds that of the nitrogen group strongly conflicts with the researchers' conclusion that nitrogen fertilizer produces larger yields. Choice D is incorrect. Although it is true that the bar graph shows a higher yield for the nitrogen group in 2018 than in 2017, without the control to compare against, it is impossible to know whether the increase is due to the fertilizer and not, for example, more favorable weather in 2018 than in 2017.

Q10

C

- C. Although Ngũgĩ's decision to write in Gĩkũyũ was met with some skepticism, it didn't prevent him from achieving literary success.

Choice C is the best answer because it most accurately states the main idea of the text. The text begins by explaining that Kenyan writer Ngũgĩ wa Thiong'o had published four novels in English before he began writing in his native language, Gĩkũyũ, in the 1970s. The text then explains that fellow writers and critics were skeptical of this decision, warning that the choice to publish in his native language, which the text implies isn't widely understood, would limit his readership. The text concludes by stating that despite those warnings, Ngũgĩ proceeded to publish several acclaimed novels in Gĩkũyũ that were translated into several languages, suggesting that Ngũgĩ's works in Gĩkũyũ achieved literary success in the forms of critical recognition and a wide international readership. Thus, the main idea is that although Ngũgĩ's decision to write in Gĩkũyũ was met with some skepticism, this choice didn't prevent him from achieving literary success.

Choice A is incorrect because it contradicts the description of Gĩkũyũ provided in the text. Rather than indicating that Gĩkũyũ is a widely understood language, as this choice claims, the text states that colleagues cautioned Ngũgĩ that writing in Gĩkũyũ instead of English "might make his works inaccessible outside his own community." Furthermore, the text contrasts the choice of Gĩkũyũ with another alternative to English, Kiswahili, which the text indicates is "a more practical choice" given that it is "widely spoken in Kenya and elsewhere in Africa." Choice B is incorrect because it contradicts information in the text. Although the text states that the novels Ngũgĩ published in Gĩkũyũ were later translated into other languages, the text explicitly states that Ngũgĩ had previously published four novels in English before he began writing in Gĩkũyũ in the 1970s. Thus, his first works were published in English, not Gĩkũyũ. Choice D is incorrect. The text doesn't address what motivated Ngũgĩ's choice to publish in his native Gĩkũyũ, and nothing in the text suggests that he intended for other authors to follow his example.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank